

ECOSYSTEM SURVIVAL



- Cristian Castañeda
- Julian Carvajal
- Juan Pablo Angulo

GAME CONCEPT AND GOAL

OBJECTIVE: REACH SAFE ZONES (X)



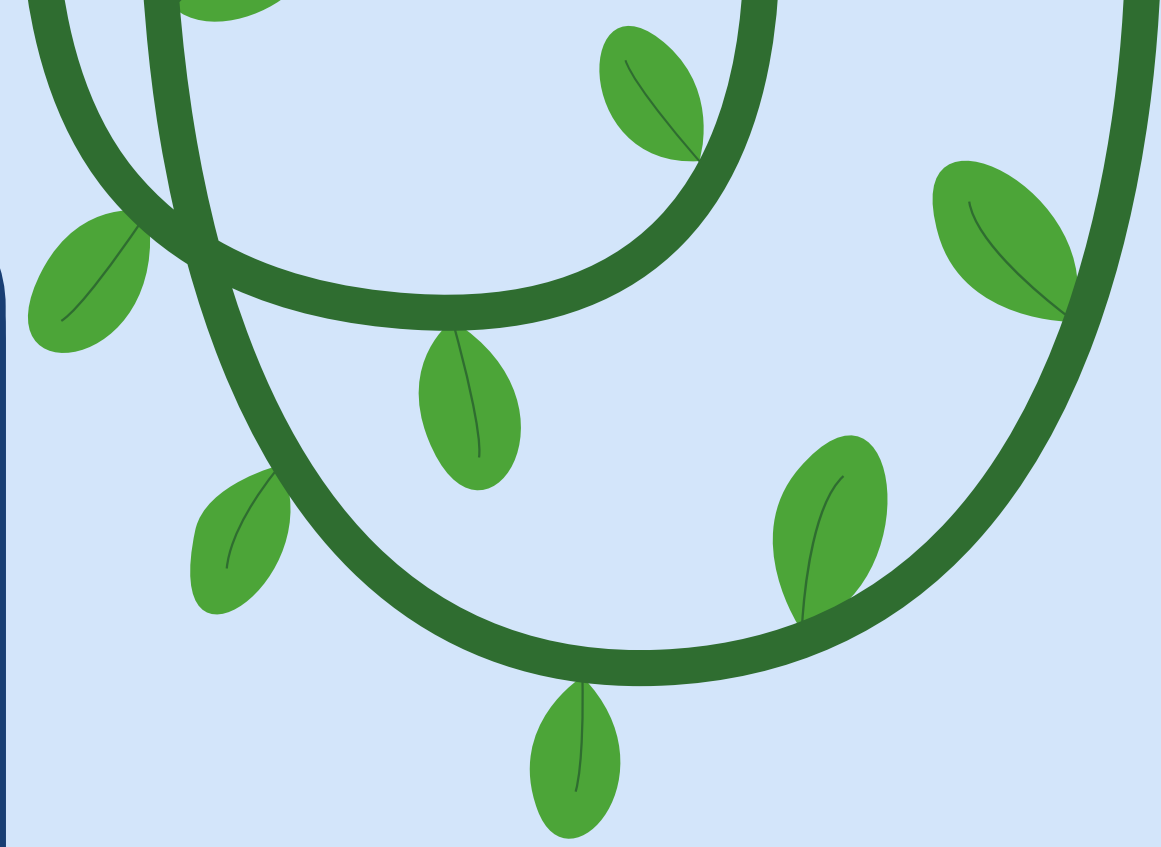
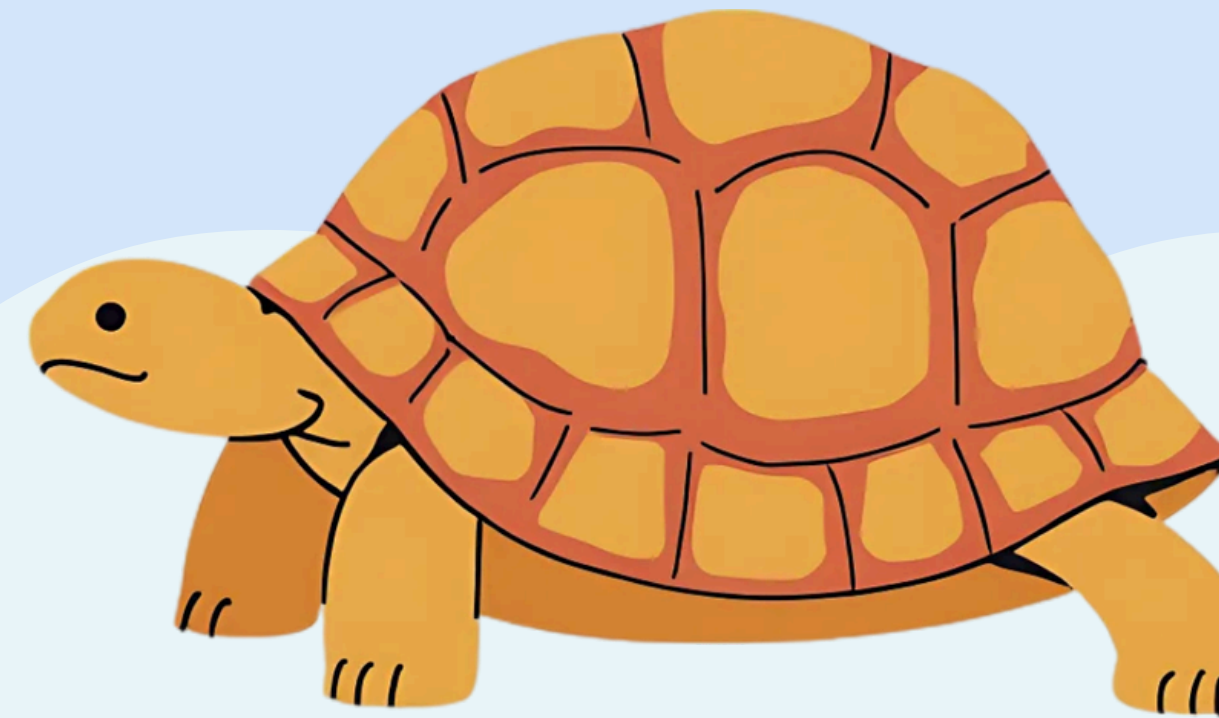
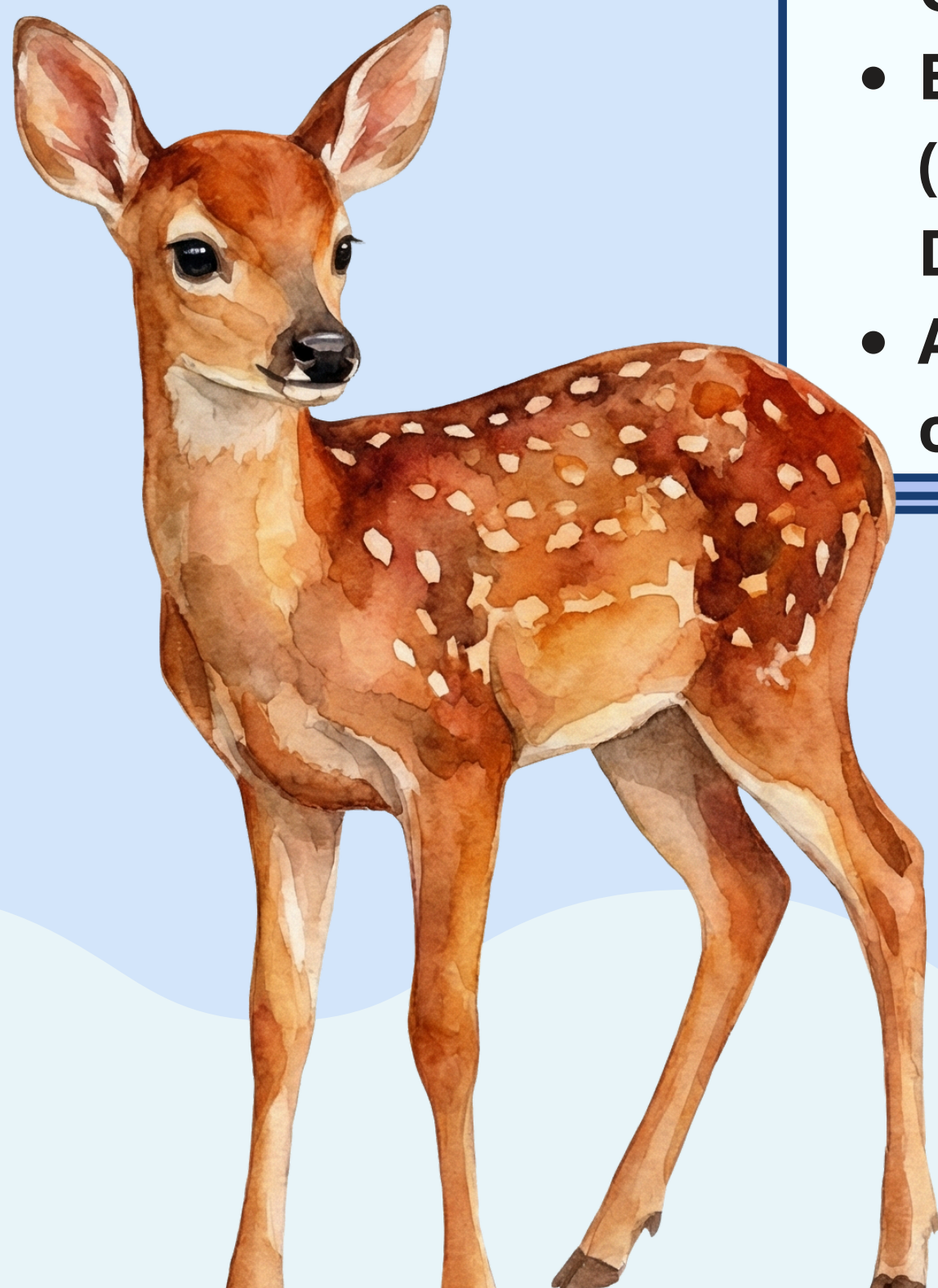
FOREST ECOSYSTEM SURVIVAL

LOSS CONDITION: SPECIES EXTINCTION (POPULATION = 0)



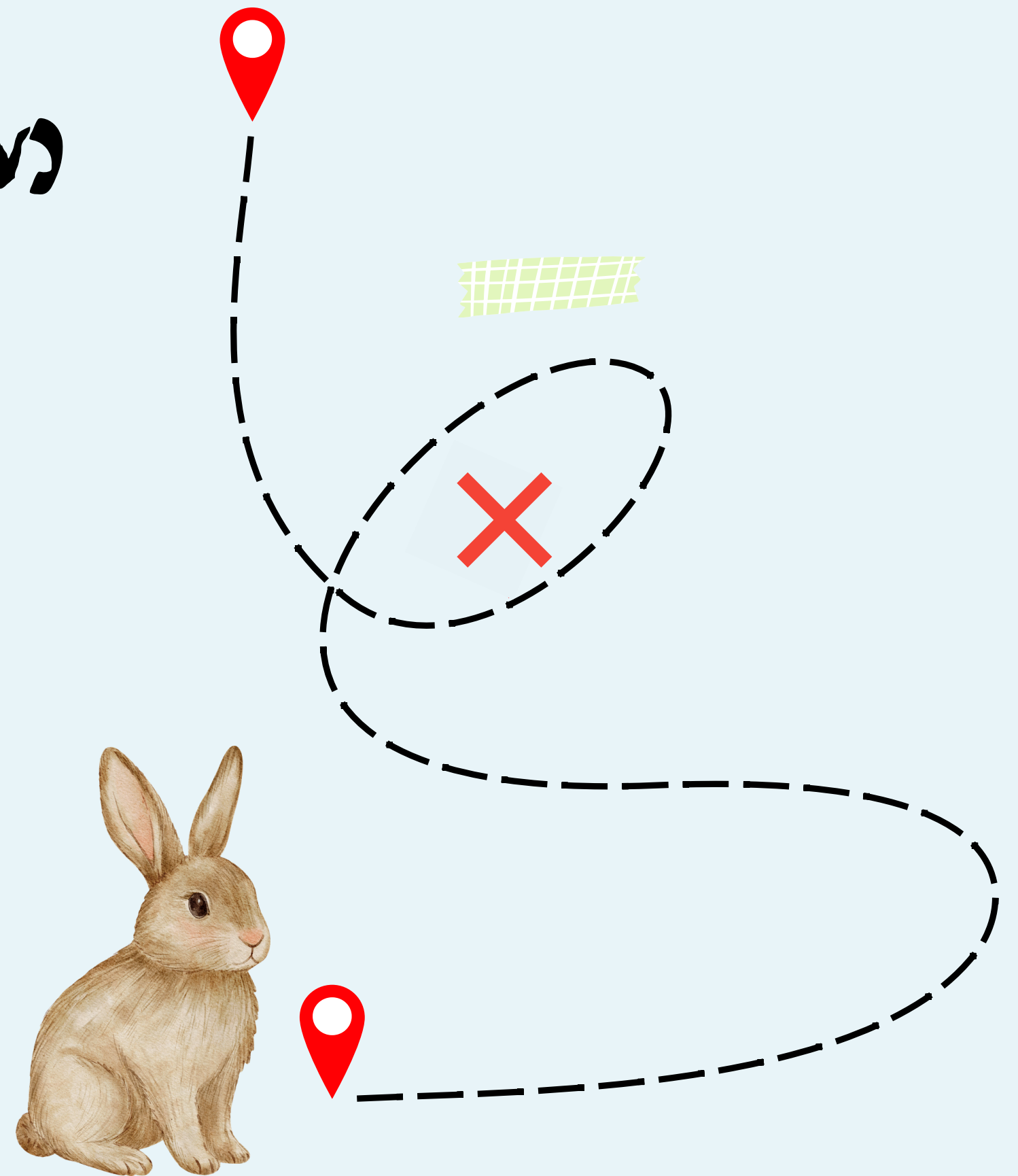
SYSTEM CONTEXT

- **Environment: 8X8 Forest Grid**
- **Entities: 4 Species (Rabbit, Fox, Turtle, Deer)**
- **Actions: 2 per turn (Feed or Migrate)**



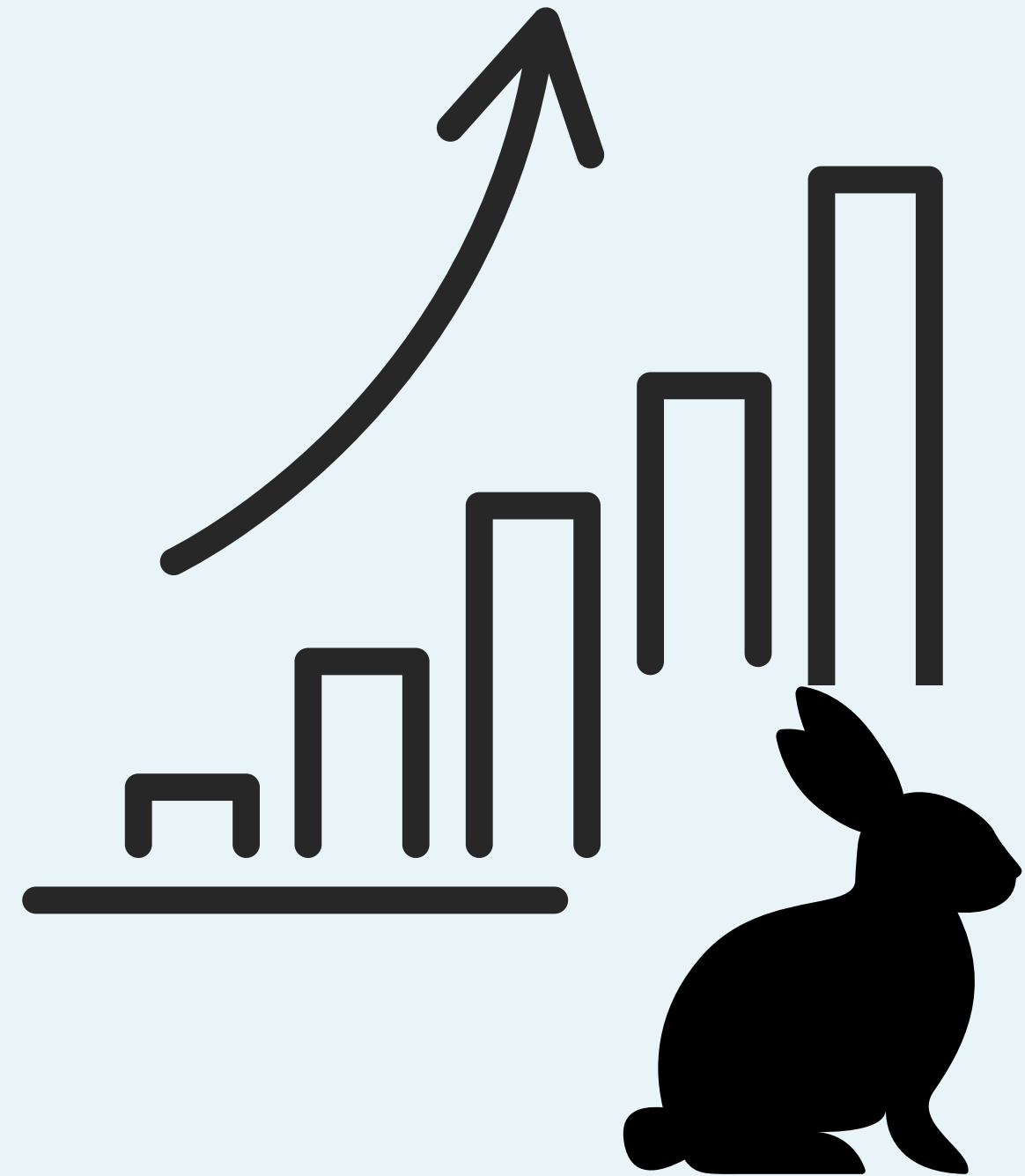
ALGORITHMIC ENGINE (LOGIC FLOW)

- **GREEDY ADVISOR: HIGH-RISK SPECIES RECOMMENDATION**
- **BACKTRACKING PATH: SAFE ROUTE CALCULATION**
- **FEEDBACK LOOP: ALGORITHMS DRIVE GAME DECISIONS**



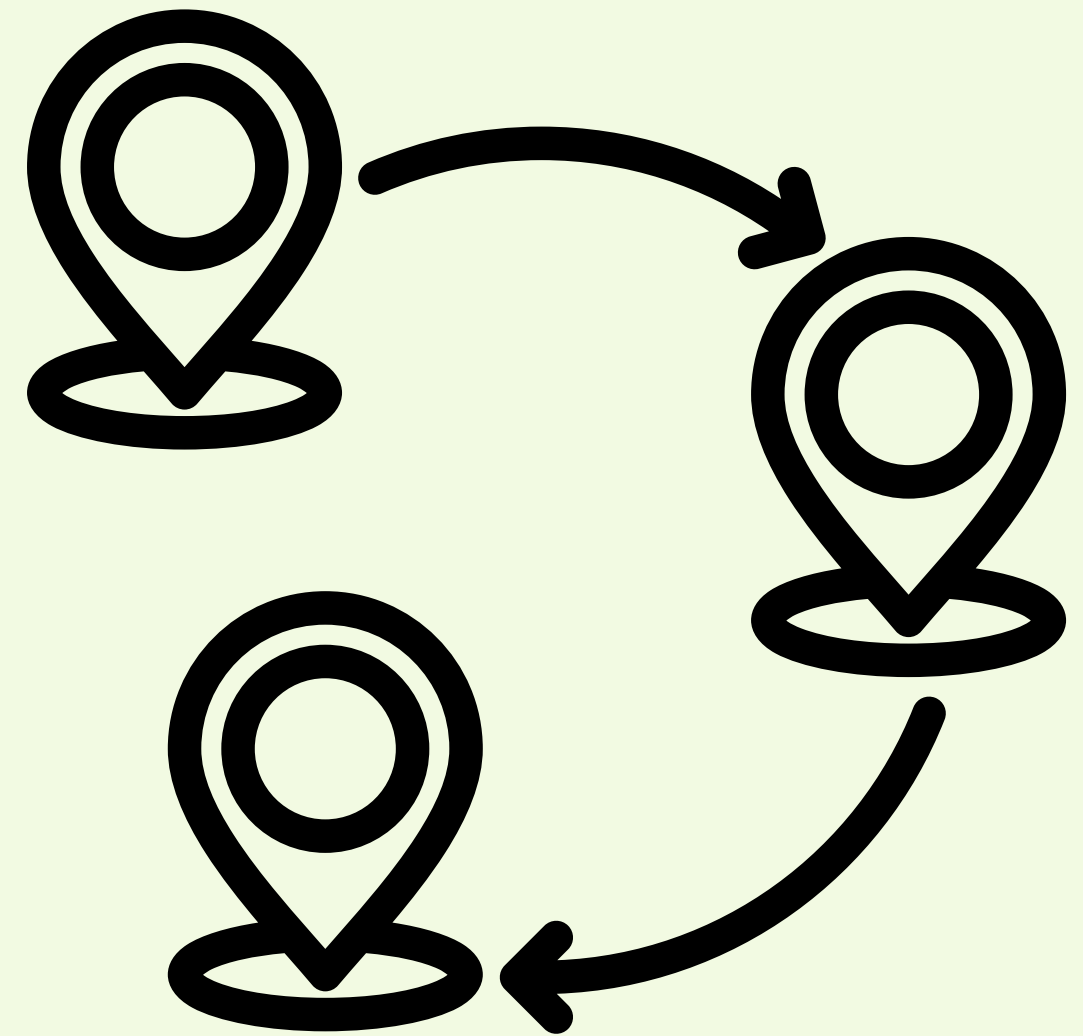
GREEDY STRATEGY

THE GREEDY ALGORITHM FUNCTIONS AS A REAL-TIME SURVIVAL ADVISOR THAT PERFORMS LOCAL OPTIMIZATION TO MANAGE SPECIES VULNERABILITY. BY ANALYZING A HEALTH MATRIX THAT INCLUDES CURRENT HUNGER LEVELS AND POPULATION COUNTS



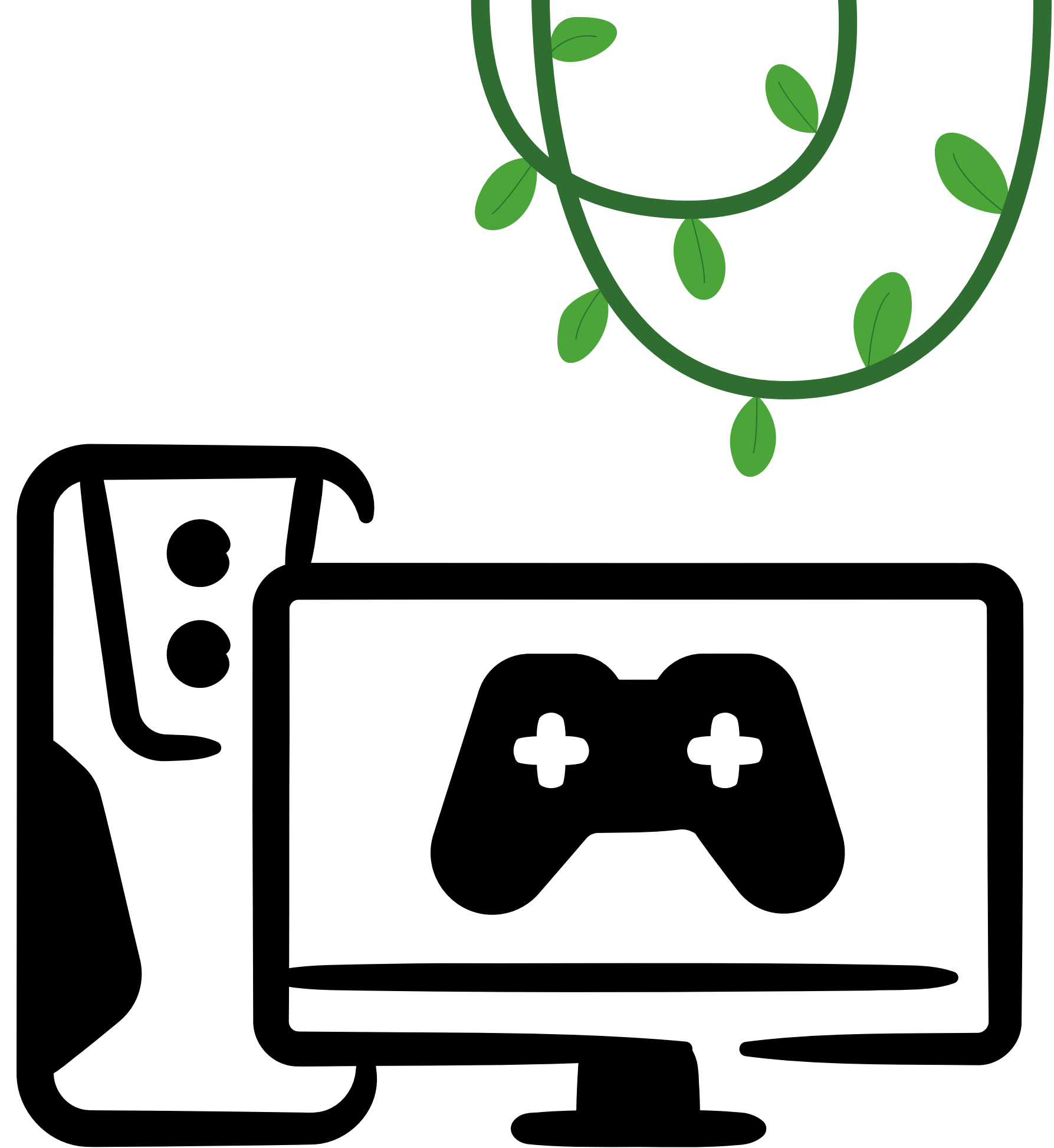
BACKTRACKING: SAFE MIGRATION PATH

BACKTRACKING IS UTILIZED AS A RECURSIVE NAVIGATION ENGINE TO FIND SAFE MIGRATION PATHS TOWARD REFUGE ZONES, EXPLORING THE 8X8 GRID SIMULATING MOVEMENT WHILE AVOID PREDATORS AND OBSTACLES.



DIGITAL ECOSYSTEM

THE GAME FUNCTIONS AS A SYNCHRONIZED SIMULATION WHERE EVERY DECISION TRIGGERS A CHAIN REACTION ACROSS THE FOREST, PYTHON ACTS AS THE CONTROLLER WHILE C++ PROVIDE THE LAWS, AND THE JSON BRIDGE IS THE INVISIBLE TRANSLATOR THAT ENSURES YOUR ACTIONS ARE INSTANTLY REFLECTED IN A STABLE, PREDICTABLE, AND FAIR GAME WORLD.



THANKS!

